THE UNIVERSITY OF MELBOURNE

Student number

Assignment 4 Semester 1, 2023

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00, AEST), Thursday 6 April 2023

This assignment consists of 5 pages (including this page)

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Instructions to Students

Writing

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the top right corner of this cover page (or the first page of your submission), and your details at the bottom of this page (or the first page).
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

Scanning

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

Submitting

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

Student Details

- Student name: _____
- Tutorial A tutor name: _____
- Tutorial A day and time (e.g. Monday 9AM): _____

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Question 1

Consider the function

$$h : \mathbb{R} \to \mathbb{R}, \qquad h(x) = x^{\frac{1}{3}} + 1 = \sqrt[3]{x} + 1.$$

(a) What is the slope of the tangent to $h(x)$ at $x = a$? Your answer should be in terms of a.

(b) What is the equation of the tangent line to $h(x)$ at the point $(a, h(a))$?
Your answer should be in terms of a.

(c) Consider the point $(0, 2)$. Does the function $h(x)$ pass through this point?

(d) Find the equation of the tangent line to $h(x)$ which passes through the point $(0, 2)$.

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Question 2

Consider the function

$$f : D \to \mathbb{R}, \qquad f(x) = \frac{1}{x^2 - 2x + 1}$$

(a) What is the maximal domain of f? Remember to show your working.

(b) Is f injective? Justify your answer.

(c) Consider another function $g : S \to \mathbb{R}$ with $g(x) = f(x)$. That is, g is defined by the same rule as f, but over a different domain, S. Find the largest set S such that g is invertible. If there are multiple options, list them, and choose S such that $2 \in S$.

(d) Given the domain S you chose, find the inverse of g and state its domain and range.

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End of Assignment